

1 MSDS: Materials Safety Data Sheets

1. standard documents that provide safety info on Materials
2. provided by suppliers
3. include info on:
 - composition
 - handling + PPE (Personal protective equipment)
 - hazards
 - shipping regulations

2 Legislation

- can support all 3 capitals
- includes state, national, and international standards+agreements

3 EV case study

People:

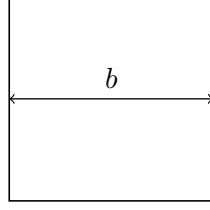
- range anxiety
-

Planet:

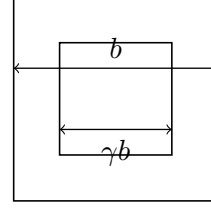
- Concerns over the carbon emission of energy grid
- are carbon emissions better for EVs?
- legislation in EU requires battery recycling

Prosperity:

- Li only produced by 2 major countries
- current Li production does not meet anticipated demand
- Nd production too low
- 95% of Nd from one country(China)



A_s (solid square)



A_h (hollow square)

Objective: Maximize EI

$$A_s = b^2$$

$$A_{\text{gap}} = (b - 2t)^2 \quad \text{where } t \text{ is thickness}$$

Thin-wall assumption: $t \ll b$

$$A_s - A_g = A_h = 4bt$$

Expanding I :

$$I = I_s - I_{\text{gap}} = \frac{b^4}{12} - \frac{(b - 2t)^4}{12} = \frac{A_s^2 - A_{\text{gap}}^2}{12} = \frac{(A_s - A_g)(A_s + A_g)}{12}$$

$$I = \frac{[4bt] (b^2 + (b - 2t)^2)}{12} = \frac{8b^3t - 16b^2t^2 + 16bt^3}{12}$$

By the thin-wall assumption: $bt^3, b^2t^2 \ll b^3t$

$$I \approx \frac{2}{3}b^3t$$

Mass constraint:

$$m = 4bt\rho L \quad \Rightarrow \quad t = \frac{m}{4b\rho L}$$

Hence

$$I \approx \frac{2}{3}b^3 \left(\frac{m}{4b\rho L} \right) = \frac{1}{6} \frac{m b^2}{\rho L}$$

Back to the objective:

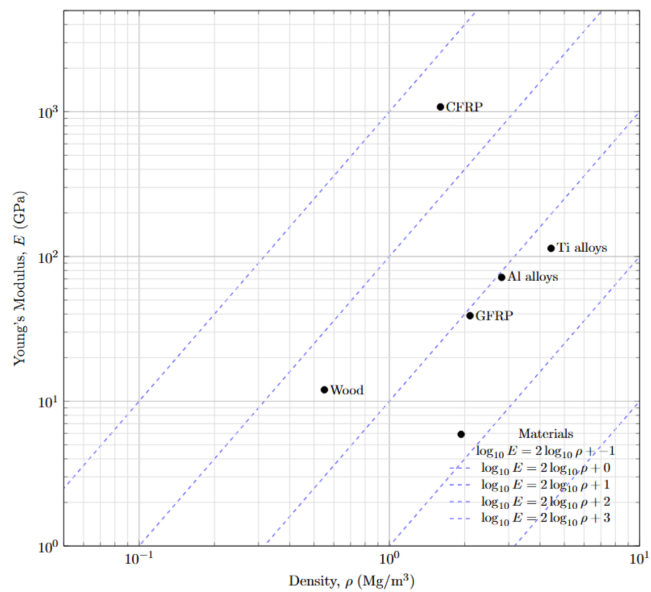
$$EI \approx E \left(\frac{2}{3}b^3 \right) \left(\frac{m}{4b\rho L} \right)$$

Define the material index

$$M = \frac{E}{\rho}$$

Taking logs for selection charts:

$$\log E = \log \rho + \log M$$



<https://hal.science/jpa-00251707/document>